Bounded-Authority Federated Recovery of Cyber–Physical Systems under Joint Physical-Channel Corruption and Recovery-Knowledge Poisoning

Kai Zhong, Zhibang Yang, Yumeng Liu, Siyang Yu, Kenli Li, *Senior Member, IEEE*

***Abstract*— Federated recovery introduces an integrity dependency that is largely absent from conventional cyber–physical system (CPS) recovery formulations: a controller may need to restore a plant while its observations and control channel are corrupted, yet the distributed recovery knowledge used to adapt the controller may itself be stale, inconsistent, or adversarially poisoned. We formulate this problem as *bounded-authority federated recovery*. Instead of treating federated knowledge as trusted state, AEFC treats it as untrusted evidence whose influence is explicitly bounded along three security-critical transitions: evidence-to-belief, belief-to-action authority, and accepted evidence-to-deployed-policy adaptation. AEFC combines a provenance-aware temporal belief update, a risk-aware recovery gate, trust-gated online policy adaptation, Byzantine-resilient federated coordination, and a robust physical execution shield. Under bounded physical corruption, bounded robust-aggregation error, contractive belief dynamics, bounded adaptation steps, uncertainty-set coverage, and shield feasibility, we establish bounded belief distortion, bounded false authorization, bounded deployed-policy deviation, finite ε-correction after poisoning under an explicit recovery condition, and policy-independent forward invariance of the physical safe set. Heterogeneous local adapters and deterministic Top-k communication are retained as implementation mechanisms rather than security roots. In an orientation-safe native MathWorks R2024b IEEE 39-bus evaluation with seven methods, four corruption cells, and 20 matched seeds per cell, all methods remain free of episode-level safety violations under joint corruption, while AEFC reduces maximum belief distortion from 0.4978 to 0.0873 and maximum deployed-policy drift from 0.1407 to 0.0230 relative to FedRL; both paired comparisons remain significant after Holm correction ($p_{\mathrm{adj}} = 1.14 \times 10^{-5}$). All numerical claims are generated from committed configurations and immutable per-seed traces.**

***Index Terms*—Cyber-physical systems, resilient recovery, federated reinforcement learning, Byzantine robustness, recovery integrity, control barrier function, trustworthy learning.**

I. INTRODUCTION

Kai Zhong and Kenli Li are with the College of Computer Science and Electronic Engineering, Hunan University, and the Ministry of Education Key Laboratory of Fusion Computing of Supercomputing and Artificial Intelligence, Changsha 410082, China.

Zhibang Yang is with the Hunan Province Key Laboratory of Industrial Internet Technology and Security, Changsha University, Changsha 410022, China.

Yumeng Liu is with the Beijing Key Laboratory of Human-Computer Interaction, Institute of Software, Chinese Academy of Sciences, Beijing 100190, China.

Siyang Yu is with the College of Information Technology and Management, Hunan University of Finance and Economics, Changsha 410000, China.

E-mail: startkz@hnu.edu.cn, yangzhibang2006@126.com, yumeng@iscas.ac.cn, yusiyang@hufe.edu.cn, lkl@hnu.edu.cn.

CYBER–PHYSICAL systems (CPS) integrate sensing, communication, computation, and control with physical processes. During post-attack recovery, these layers cannot always be restored sequentially. A recovery controller may have to act while observations remain partially corrupted, the communication path is delayed or replayed, and the command delivered to the actuator is perturbed. When recovery is further adapted through federated reinforcement learning (FRL), a second integrity dependency appears: the distributed knowledge used to update the recovery policy can itself be Byzantine, stale, semantically incompatible, or strategically poisoned.

This combination creates a *recovery-integrity* problem. The question is no longer only whether a learned controller can recover the plant efficiently, but whether corrupted recovery knowledge can acquire enough authority to distort the controller's belief, authorize a risky action, alter the deployed policy, and finally affect the physical plant. Standard FRL does not answer this question. Data locality does not prevent model poisoning, and a robust aggregation rule that reduces optimization error does not imply that the resulting physical action is safe. Conversely, placing a safety filter after an arbitrary learned policy can protect the plant, but by itself it does not explain how malicious recovery knowledge is prevented from persistently controlling the learning and authorization process.

This paper therefore asks the following research question:

> *Can a federated recovery controller remain physically safe and retain bounded recovery authority when both the physical observation/control channels and the federated recovery knowledge are adversarially corrupted?*

Our answer is conditional but explicit: yes, when physical corruption remains inside a modeled attack envelope, the uncertainty set covers the true state, the influence of corrupted federated knowledge is bounded, the belief update is contractive, online adaptation is rate-limited and gated, and the robust execution shield remains feasible. These conditions make the security boundary visible rather than hiding it inside empirical recovery accuracy.

We instantiate this principle in AEFC. Heterogeneous clients keep subsystem-specific observation and action adapters local and federate only a representation-compatible recovery backbone. However, heterogeneity handling is not the security root. The security path instead consists of four layers.

First, provenance-aware temporal processing converts potentially corrupted local and federated evidence into a bounded recovery belief. Second, a risk-aware gate determines whether a learned proposal receives recovery authority. Third, trust-gated online adaptation separates candidate updates from the deployed controller and bounds the amount by which accepted evidence can move the deployed policy. Finally, a robust CBF/CLF-based execution shield enforces the hard physical safety contract before the command is delivered to the plant.

The execution order is important. Let u_t^{π} denote the learned proposal and $u_t^{\star}$ the shielded command. We model control-channel corruption *after* the shield and before the plant:

$$u_t^{\mathrm{exec}} = u_t^{\star} + a_t^{u}. \tag{1}$$

This placement prevents an artificial guarantee in which the shield is allowed to "correct" an attack that in reality occurs downstream of the controller. Safety is therefore claimed only for a_t^u inside an explicitly modeled admissible set.

The resulting architecture also clarifies the role of federated coordination. Certificate-supported local transitions, representation alignment, Top-k sparsification, and Byzantine-resilient aggregation improve the quality and efficiency of distributed learning. They do not authenticate a Byzantine client and they do not replace the physical safety layer. Client-reported advantage, prototype similarity, or confidence is never treated as a trust root. The server limits Byzantine influence with a mechanism whose assumptions are stated explicitly, while the deployed controller limits the authority of accepted knowledge before physical execution.

The main contributions are as follows.

- We formulate *bounded-authority federated recovery* under joint physical-channel corruption and recovery-knowledge poisoning. The formulation separates evidence integrity, recovery authorization, online adaptation, and physical execution rather than collapsing them into one empirical robustness score.
- We develop a provenance-aware temporal belief update, a risk-aware recovery gate, and trust-gated online policy adaptation. Together they bound the propagation of corrupted recovery knowledge from evidence to belief, from belief to action authority, and from accepted updates to the deployed policy.
- We provide a compositional security analysis. The guarantees include bounded belief distortion, bounded false authorization, bounded deployed-policy deviation, finite ε-correction after poisoning under an explicit post-attack contraction condition, and policy-independent forward invariance under bounded post-shield control corruption.
- We retain heterogeneous adapters, deterministic Top-k compression, and Byzantine-resilient coordination as implementation components, but revise their claims to match their actual security role. In particular, sparsification is treated as communication optimization rather than a privacy or trust mechanism.
- We evaluate the composition under joint physical/knowledge corruption using matched seeds, per-seed raw traces, security-specific metrics, paired statistical tests, and explicit reporting of the theorem boundary. Numerical claims that are not traceable to executable configurations and raw logs are excluded from the reported evidence.

The remainder of this paper reviews related work, presents the system and threat model and AEFC mechanisms, develops the security analysis, reports the experimental and validation results, and concludes with the scope and limitations of the guarantees.

II. RELATED WORK

A. Resilient Recovery and Secure Control of CPS

CPS recovery has been studied using resilient estimation, robust control, optimization, game-theoretic methods, and learning-based decision making [1]–[9]. These approaches establish important recovery principles, but the information consumed by the recovery controller is frequently assumed to be sufficiently trustworthy. Our setting is different because sensing, control delivery, and the distributed learning process that adapts the controller may be attacked concurrently.

A key distinction is between learning robustness and physical safety. Even a statistically robust global model can produce an unsafe action in a safety-critical plant, while a model-independent safety constraint can preserve a physical invariant even when the performance controller is poor. We therefore use the learned policy only as a recovery proposal and place a robust execution contract at the physical authority boundary.

B. Reinforcement Learning for CPS Recovery

RL and multi-agent RL have been used for sequential recovery and distributed CPS control [10]–[18]. These methods motivate the actor–critic learner used by AEFC. However, a critic-estimated advantage is only a utility signal. Under corrupted observations or a poisoned critic, a large estimated advantage does not certify state correctness or physical safety. AEFC therefore separates recovery utility from authorization and execution safety.

C. Federated Learning under Heterogeneity and Communication Constraints

Federated learning reduces raw-data exchange but is affected by non-IID data, client drift, system heterogeneity, communication cost, and malicious updates [19], [20]. Personalized and representation-aligned methods support decompositions in which a shared model is combined with client-specific components [21]–[24]. AEFC follows this pattern by federating a recovery backbone while keeping incompatible observation and action adapters local.

Communication-efficient FL often applies sparsification or compressed updates [25], [26]. Compression does not authenticate a client and does not provide formal privacy. We therefore analyze deterministic Top-k only as a communication mechanism and keep its residual explicit.

D. Poisoning and Byzantine-Resilient Federated Learning

Secure federated learning studies poisoning, backdoors, malicious clients, client screening, robust aggregation, secure computation, trusted execution, and differential privacy [27]–[34]. Federated RL has also been investigated for distributed control and cyber defense [35]–[39]. These studies demonstrate that federated coordination can remain useful under adversarial clients, but aggregation integrity alone does not specify how much physical authority a poisoned recovery model should receive.

E. Gap: Recovery-Knowledge Authority

The security gap addressed here lies in the propagation path from distributed evidence to the plant. Robust aggregation can bound a model update, and a robust safety filter can bound the final action, but the intermediate transitions—belief formation, action authorization, and online deployment—remain security-relevant. AEFC therefore treats federated recovery knowledge as untrusted evidence with *bounded authority*. This composition, rather than Top-k, advantage estimation, prototype alignment, or a federated critic individually, is the security focus of the paper.

III. METHODOLOGY

A. System Model and Recovery Objective

We consider a CPS partitioned into N interacting subsystems. Subsystem i has state $x_i \in \mathbb{R}^{n_i}$ and control input $u_i \in \mathbb{R}^{m_i}$. Its attacked dynamics are

$$\dot{x}_i = f_i(x_i, x_{-i}) + g_i(x_i)(u_i + a_i^u) + w_i, \tag{2}$$

and the received observation is

$$y_i = \mathcal{H}_i(x_i) + a_i^y, \tag{3}$$

where $\mathcal{H}_i$ is the observation map, a_i^y models observation/communication corruption, a_i^u models corruption between the shield output and the plant input, and w_i represents bounded modeling disturbance.

Let

$$\mathcal{S}_i = \{x_i : h_{i,\ell}(x_i) \geq 0,\ \ell = 1, \ldots, M_i\} \tag{4}$$

denote the physical safe set and

$$\mathcal{R}_i = \{x_i : V_i(x_i) \leq \varepsilon_i\} \tag{5}$$

a recovery set defined by a Lyapunov candidate V_i. Each subsystem maintains a state uncertainty set $\widehat{\mathcal{X}}_i(t)$ and admissible attack/disturbance sets $\mathcal{A}_i^u(t)$ and $\mathcal{W}_i(t)$. The physical guarantees below are conditional on

$$x_i(t) \in \widehat{\mathcal{X}}_i(t), \quad a_i^u(t) \in \mathcal{A}_i^u(t), \quad w_i(t) \in \mathcal{W}_i(t). \tag{6}$$

B. Threat Model and Trust Boundary

The adversary may act at two layers.

Physical-channel adversary. The attacker can bias, replay, delay, or partially suppress observations and can perturb the command after it leaves the controller, subject to the modeled attack envelope. An attacker that can impose an unbounded arbitrary actuator command is outside the guarantee.

Federated recovery-knowledge adversary. At most f of N participating clients are Byzantine. Byzantine clients may collude and submit arbitrary update values, sparse supports, timestamps, prototypes, advantages, confidence values, or other self-reported metadata. They may also replay stale but syntactically valid recovery knowledge. The server follows the coordination protocol.

Authenticated provenance fields and client self-reported semantic fields are conceptually distinct. When cryptographic source authentication, secure time, or remote attestation is unavailable, the corresponding provenance property must be treated as an implementation assumption rather than inferred from a client-provided identifier.

The central security requirement is:

> Corrupted federated knowledge may affect recovery utility, but its influence on belief, authorization, deployed-policy adaptation, and physical actuation must remain bounded by explicit mechanisms and assumptions.

C. Heterogeneous Local Recovery Learner

Client i uses

$$\pi_i(o_i) = g_{\xi_i}(b_\theta(e_{\psi_i}(o_i)))\,, \tag{7}$$

where e_{ψ_i} is a local observation adapter, b_θ is the shared recovery backbone, and g_{ξ_i} is a local action adapter. Only θ is federated. This design prevents direct averaging of incompatible subsystem input/output layers.

Let $Q_i(s, a)$ be the local recovery critic. The recovery advantage

$$A_i(s, a) = Q_i(s, a) - Q_i(s, \pi(s)) \tag{8}$$

is used only as a utility signal and never as a security credential.

D. Provenance-Aware Temporal Belief Update

Let evidence from source j at time t be

$$e_{j,t} = (z_{j,t}, p_{j,t}), \tag{9}$$

where $z_{j,t}$ is the recovery evidence and

$$p_{j,t} = (\mathrm{src}_j, \tau_{j,t}, v_{j,t}^{\mathrm{model}}, \ell_{j,t}) \tag{10}$$

contains provenance attributes such as source, timestamp, model version, and lineage.

Let $\chi_{j,t} \in \{0, 1\}$ denote provenance validity under the deployed authenticity mechanism, $\Delta_{j,t} = t - \tau_{j,t}$ the evidence age, and

$$r_{j,t} = d\Big(z_{j,t}, \mathcal{C}(b_{t-1}, \widehat{\mathcal{X}}_t)\Big) \tag{11}$$

a physics/temporal consistency residual. We define a bounded influence weight

$$\omega_{j,t} = \chi_{j,t} e^{-\beta \Delta_{j,t}} \psi_c(r_{j,t}), \qquad 0 \le \psi_c(\cdot) \le 1. \tag{12}$$

A robust recovery-knowledge summary is

$$\bar{k}_t = \mathcal{R}_f(\{\omega_{j,t} z_{j,t}\}_{j=1}^N), \tag{13}$$

and the recovery belief is updated by

$$b_{t+1} = \Pi_{\mathcal{B}} T(b_t, \widehat{y}_t, \bar{k}_t). \tag{14}$$

We take $\mathcal{B}$ to be closed and convex and $\Pi_{\mathcal{B}}$ to be the Euclidean projection; hence the projection used in the analysis is non-expansive.

Assumption 1 (Contractive recovery-belief dynamics)**.** *There exist $0 \le \lambda_b < 1$, $L_y \ge 0$, and $L_k \ge 0$ such that*

$$\|T(b,y,k) - T(b',y',k')\| \le \lambda_b \|b - b'\| + L_y \|y - y'\| + L_k \|k - k'\|. \tag{15}$$

Theorem 1 (Bounded belief distortion)**.** *Let b_t° denote the clean counterfactual belief. If $\|a_t^y\| \le \epsilon_y$ and $\|\bar{k}_t - \bar{k}_t^\circ\| \le \epsilon_k$ for all t, then*

$$\|b_t - b_t^\circ\| \le \lambda_b^t \|b_0 - b_0^\circ\| + \frac{L_y \epsilon_y + L_k \epsilon_k}{1 - \lambda_b} (1 - \lambda_b^t). \tag{16}$$

Proof. Projection onto $\mathcal{B}$ is non-expansive. Applying the Lipschitz condition to the attacked and clean recursions gives $e_{t+1} \le \lambda_b e_t + L_y \epsilon_y + L_k \epsilon_k$. Unrolling the recursion yields (16). □

If poisoning stops at T_p and the combined post-attack discrepancy entering the belief recursion is bounded by ϵ_{clean}, the same recursion gives geometric correction toward an $\epsilon_{\text{clean}}/(1 - \lambda_b)$ residual neighborhood. Exact finite-time return is not claimed without zero residual disturbance or an additional reset/rollback mechanism.

E. Certificate-Supported Local Recovery Knowledge

For an honest client, a local transition $\tau = (s, a, r, s')$ is eligible for federation only when it is recovery-useful and consistent with local safety/recovery evidence. Define

$$m_i^{\text{safe}}(\tau) = \min_{\ell} \inf_{\substack{x \in \widehat{\mathcal{X}}_i \\ a^u \in \mathcal{A}_i^u \\ w \in \mathcal{W}_i}} \Big[\nabla h_{i,\ell}^\top (f_i + g_i(u + a^u) + w) + \alpha_\ell(h_{i,\ell}) \Big] \tag{17}$$

and

$$m_i^{\text{rec}}(\tau) = - \sup_{\substack{x \in \widehat{\mathcal{X}}_i \\ a^u \in \mathcal{A}_i^u \\ w \in \mathcal{W}_i}} [\dot{V}_i(x, u, a^u, w) + c_i V_i(x)]. \tag{18}$$

The honest-client sharing set is

$$\mathcal{B}_i^+ = \{\tau : A_i(\tau) > \zeta_A,\ m_i^{\text{safe}}(\tau) \ge 0,\ m_i^{\text{rec}}(\tau) > \zeta_V\}. \tag{19}$$

This gate limits what an honest client shares. It does not authenticate Byzantine certificates.

F. Representation Alignment and Sparse Coordination

For coarse recovery mode m, honest client i may construct a latent prototype

$$p_{i,m} = \frac{1}{|\mathcal{B}_{i,m}|} \sum_{o \in \mathcal{B}_{i,m}} e_{\psi_i}(o) \tag{20}$$

and use a robust center $\bar{p}_m$ only to reduce honest-client representation drift.

For communication, let $C_k(v)$ retain the k largest-magnitude coordinates. With error feedback,

$$z_i^t = g_i^t + e_i^t, \tag{21}$$

$$U_i^t = C_k(z_i^t), \tag{22}$$

$$e_i^{t+1} = z_i^t - U_i^t. \tag{23}$$

Lemma 1 (Deterministic Top-k contraction)**.** *For any $v \in \mathbb{R}^d$,*

$$\|v - C_k(v)\|_2^2 \le \left(1 - \frac{k}{d}\right) \|v\|_2^2. \tag{24}$$

The contraction result does not justify treating an untransmitted coordinate as an observed numerical zero in a Byzantine proof. To keep the aggregation semantics well-defined, the preferred implementation uses a common support S_t, $|S_t| = k$, determined independently of any single client's report; all participating clients then upload $g_{i,t}[S_t]$. If client-specific supports are used instead, the server must employ a support-aware robust estimator and the corresponding theorem must be changed accordingly.

On a common coordinate population, the server applies a Byzantine-resilient estimator $\mathcal{R}_f$. For coordinate-wise f-trimmed mean, if all honest values on a coordinate lie in $[\mu_{t,j}^H - r_{t,j}, \mu_{t,j}^H + r_{t,j}]$ and $f < N/2$, then

$$|\widehat{g}_{t,j} - \mu_{t,j}^H| \le r_{t,j}. \tag{25}$$

This envelope is the source of ϵ_k used in the belief analysis; client-reported confidence or advantage is not.

G. Risk-Aware Recovery Gate

The belief and learned policy produce a candidate u_t^π. Before this proposal is allowed to exercise recovery authority, AEFC computes a robust safety shortfall

$$R_t^{\text{safe}}(u) = \max_{\ell} \sup_{\substack{x \in \mathcal{X}(b_t) \\ a^u \in \mathcal{A}_t^u \\ w \in \mathcal{W}_t}} [-\mathcal{B}_\ell(x, u, a^u, w)]_+, \tag{26}$$

where

$$\mathcal{B}_\ell(x, u, a^u, w) = \nabla h_\ell(x)^\top [f(x) + g(x)(u + a^u) + w] + \alpha_\ell(h_\ell(x)). \tag{27}$$

We combine this term with recovery and uncertainty penalties:

$$\widehat{\mathfrak{R}}_t(u) = \lambda_s R_t^{\text{safe}}(u) + \lambda_r [-m_t^{\text{rec}}(u)]_+ + \lambda_b \operatorname{diam}(\mathcal{X}(b_t)) + \lambda_p (1 - q_t), \tag{28}$$

where $q_t \in [0, 1]$ is a protocol-level trust state derived from provenance validity, age, cross-source consistency, and belief uncertainty.

The learned proposal is authorized only when

$$G_t(u) = \mathbf{1}\{\widehat{\mathfrak{R}}_t(u) \le \rho_{\max} - \epsilon_R\}\mathbf{1}\{q_t \ge q_{\min}\}. \tag{29}$$

Then

$$u_t^{\text{prop}} = \begin{cases} u_t^{\pi}, & G_t = 1, \\ u_t^{\text{fallback}}, & G_t = 0. \end{cases} \tag{30}$$

Theorem 2 (Bounded false authorization)**.** *Assume*

$$\Pr[\mathfrak{R}_t^{\text{true}}(u) \le \widehat{\mathfrak{R}}_t(u) + \epsilon_R] \ge 1 - \delta_g. \tag{31}$$

Then

$$\Pr[G_t(u) = 1 \ \wedge \ \mathfrak{R}_t^{\text{true}}(u) > \rho_{\max}] \le \delta_g. \tag{32}$$

If all uncertainty sets are deterministic outer approximations, δ_g can be zero.

H. Trust-Gated Online Policy Adaptation

AEFC separates a candidate model from the deployed controller. Given robust aggregate g_t, the candidate is

$$\widetilde{\theta}_{t+1} = \Pi_{\mathbb{B}(\theta_t^{\text{dep}}, \Delta_\theta)}\left[\theta_t^{\text{dep}} - \eta_t q_t \ \text{clip}_C(g_t)\right]. \tag{33}$$

Deployment requires

$$A_t = \mathbf{1}\{q_t \ge q_{\min}\}\mathbf{1}\{G_t = 1\}\mathbf{1}\{\text{validation}_t = 1\}, \tag{34}$$

and

$$\theta_{t+1}^{\text{dep}} = A_t\widetilde{\theta}_{t+1} + (1 - A_t)\theta_t^{\text{dep}}. \tag{35}$$

Theorem 3 (Bounded deployed-policy deviation)**.** *For every accepted round,*

$$\|\theta_{t+1}^{\text{dep}} - \theta_t^{\text{dep}}\| \le \min\{\eta_t C, \Delta_\theta\}. \tag{36}$$

If π_θ is L_π-Lipschitz in its parameters, then

$$\|\pi_{\theta_{t+1}^{\text{dep}}}(x) - \pi_{\theta_t^{\text{dep}}}(x)\| \le L_\pi \min\{\eta_t C, \Delta_\theta\}. \tag{37}$$

Finite correction after poisoning additionally requires either finite detection and rollback to a certified checkpoint, or a clean post-attack contraction.

Corollary 1 (Finite ε-correction after poisoning)**.** *Suppose poisoning stops at T_p and, for $t \ge T_p$,*

$$\|\theta_{t+1} - \theta^c\| \le \kappa\|\theta_t - \theta^c\| + \epsilon_c, \qquad 0 \le \kappa < 1. \tag{38}$$

Then, for every integer $s \ge 0$,

$$\|\theta_{T_p+s} - \theta^c\| \le \kappa^s\|\theta_{T_p} - \theta^c\| + \frac{1 - \kappa^s}{1 - \kappa}\epsilon_c. \tag{39}$$

Consequently, for any $\varepsilon > \epsilon_c/(1-\kappa)$, the deployed parameters enter the ε-ball around θ^c after finitely many clean rounds. Exact return requires $\epsilon_c = 0$ or a certified rollback/reset mechanism.

Proof. Unrolling the contraction recursion for s clean rounds gives the stated geometric bound. Since $\kappa^s \to 0$, every radius strictly larger than the residual floor $\epsilon_c/(1-\kappa)$ is reached in finite time. □

I. Robust Physical Execution Shield

For compactness, let x and u denote the stacked subsystem state and input and let $\mathcal{S} = \prod_i \mathcal{S}_i$. Assume f and g are locally Lipschitz, each barrier function h_ℓ is continuously differentiable, and α_ℓ is an extended class-$\mathcal{K}$ function. The authorized or fallback proposal is still not directly applied. AEFC solves

$$u_t^\star = \arg\min_{u,\delta \ge 0} \|u - u_t^{\text{prop}}\|_2^2 + \lambda_\delta \delta^2 \tag{40}$$

subject to the hard robust safety constraints

$$\inf_{\substack{x \in \widehat{\mathcal{X}}_t \\ a^u \in \mathcal{A}_t^u \\ w \in \mathcal{W}_t}} \Big[\nabla h_\ell(x)^\top (f(x) + g(x)(u + a^u) + w) + \alpha_\ell(h_\ell(x))\Big] \ge 0 \tag{41}$$

and a soft recovery CLF constraint that may be relaxed by δ; the safety constraints remain hard. The actual plant receives

$$u_t^{\text{exec}} = u_t^\star + a_t^u. \tag{42}$$

Theorem 4 (Compositional safety under joint corruption)**.** *Suppose* (6) *holds for all t, the hard constraints in* (41) *are feasible, and $x(0) \in \mathcal{S}$. Then*

$$x(t) \in \mathcal{S}, \qquad \forall t \ge 0, \tag{43}$$

for any recovery belief, federated knowledge, deployed policy, or learned proposal generated under poisoning, provided the realized physical corruption remains inside the modeled envelope.

Proof. For every admissible state, downstream control corruption, and disturbance, the robust barrier constraint enforces $\dot{h}_\ell + \alpha_\ell(h_\ell) \ge 0$. Standard forward-invariance arguments therefore preserve every $\{h_\ell \ge 0\}$ and hence their intersection. The proof is independent of the quality or origin of the learned proposal. □

J. Guarantee Summary and Scope

The security guarantees are intentionally separated:

- G1: bounded belief distortion, (44)
- G2: bounded false authorization, (45)
- G3: finite ε-correction above the residual floor under explicit contraction/rollback, (46)
- G4: bounded deployed-policy deviation, (47)
- G5: physical forward invariance. (48)

G5 does not require convergence of the federated learner. G1–G4 explain why federated recovery knowledge has bounded authority rather than being made irrelevant by the final shield.

IV. EXPERIMENTAL EVALUATION

A. Evidence Policy and Evaluation Backends

Every quantitative result reported in this paper is generated from a committed executable configuration and per-seed traces. Tables and statistical summaries are derived from those

traces rather than from manually specified target curves. The released artifact contains two deliberately separated evaluation paths.

The first path is a reduced-order CPS integration backend used to exercise the complete PATBU–risk-gate–trust-gated-adaptation–aggregation–shield–post-shield-attack composition. It is reported only as mechanism-level evidence. The second path is the native MathWorks R2024b `IEEE39BusSystem` model. The native workflow materializes the official example, discovers candidate observation and control edges, validates the selected pairs by finite-difference causal probing, instruments the validated model, and executes the same method/scenario matrix used by the mechanism-level evaluation.

The authoritative native run uses four generator-local physical rotor-speed/`Pref` pairs at Gen10@Bus30, Gen1@Bus39, Gen2@Bus31, and Gen3@Bus32. The resulting 4×4 finite-difference sensitivity matrix has rank four and condition number 5.13. The native evidence is tied to commit `cc315ff522c90ce6adc09df3020472dd9ad820ea`, uses a $1/60$ s sample time, and executes $7 \times 4 \times 20 = 560$ episodes with one Fast Restart compile and numeric tunable selectors. The orientation-safe extraction path applies per-episode semantic guards before a trace is accepted.

B. Methods and Joint-Corruption Matrix

The comparison includes a standard federated RL baseline (FedRL), a robust-aggregation baseline (RobustAgg), AEFC without PATBU, AEFC without the risk gate, AEFC without trust-gated online policy adaptation, AEFC without the physical shield, and AEFC-Full. Heterogeneous adapters and representation alignment are not treated as security ablations in the reported tables because they address subsystem compatibility rather than recovery authority.

The four scenarios form a 2×2 corruption matrix:

$$\{\text{physical corruption off/on}\} \times \{\text{knowledge poisoning off/on}\}, \tag{49}$$

which yields clean, knowledge-only, physical-only, and joint conditions. The same 20 seeds, methods, and scenario labels are used across backends. In the native run, seven federated clients include two Byzantine clients. The configured attack window is steps 15–43; observation corruption is bounded to 0.25 of the normalization scale, downstream post-shield control corruption is bounded by 0.005 per channel, the model-replacement scale is 6, and stale Byzantine evidence is delayed by seven steps.

Observation attacks perturb the selected physical rotor-speed channels. Knowledge attacks include model replacement, sign-changing poisoned updates, stale evidence, and colluding Byzantine clients. Control corruption is injected after the shield and before the validated native actuator edge, preserving the threat-model ordering $u_t^{\pi} \to u_t^{\star} \to u_t^{\mathrm{exec}}$. The formal guarantee applies only inside the modeled uncertainty and attack sets; the reported experiments do not claim safety against arbitrary unbounded downstream actuation.

C. Released Evidence Contract

The native JSONL traces and per-seed summaries expose the quantities used by the manuscript. Table I lists the actual evidence schema rather than an aspirational logging interface. The mechanism-level traces additionally record the clean belief reference, robust aggregate, aggregate deviation, common support, and cumulative transmitted bytes.

TABLE I
RELEASED NATIVE EVIDENCE FIELDS USED FOR CODE-TO-CLAIM REPRODUCIBILITY.

Category	Recorded quantities
Identity	commit/configuration identifiers, mapping policy, seed, method, scenario, time
State	native and normalized observations, native safe-envelope indicator
Integrity	trust score, risk bound, gate decision, belief distortion, deployed-policy movement
Decisions	attack-active flag, dangerous-proposal flag, adaptation decision, shield intervention
Control	shield action, post-shield corruption, executed correction
Summary	safety violation, minimum native margin, FAR, maximum belief distortion, maximum policy movement, recovery success/time, intervention ratio

Separating the shield action and the post-shield corruption is essential: it distinguishes learned-policy authority, safety-layer intervention, and downstream control-channel corruption. In the native plant, these signals are combined only at the validated control edge.

D. Security and Recovery Metrics

Belief integrity is measured against the clean counterfactual belief,

$$D_b(t) = \|b_t - b_t^{\circ}\|_2. \tag{50}$$

The reported policy-drift metric is the movement of the deployed policy from its pre-update reference,

$$D_\theta(t) = \|\theta_t^{\mathrm{dep}} - \theta_0^{\mathrm{dep}}\|_2, \tag{51}$$

which matches the quantity recorded by both executable backends. It should not be interpreted as a normalized attacked-versus-clean parameter distance unless such a clean parameter trajectory is explicitly recorded.

For the native IEEE 39-bus path, let z_t denote the four normalized physical rotor-speed observations and let $\ell_{\mathrm{safe}} = 1$ be the configured normalized envelope. The instantaneous native margin is

$$m_t^{\mathrm{native}} = \ell_{\mathrm{safe}} - \|z_t\|_\infty, \tag{52}$$

and each episode reports $\min_t m_t^{\mathrm{native}}$. A negative value therefore denotes an observed envelope violation.

The false-authorization rate is computed from the logged dangerous-proposal indicator $d_t \in \{0, 1\}$:

$$\mathrm{FAR} = \frac{\sum_t \mathbf{1}\{G_t = 1 \wedge d_t = 1\}}{\max\{\sum_t d_t, 1\}}. \tag{53}$$

The indicator is backend-specific but fixed by the executable configuration: the mechanism-level path uses a one-step true-state surrogate safety criterion, whereas the native path uses

the normalized predicted envelope associated with the validated sensitivity model. Consequently, empirical FAR is an operational authorization metric; it is not itself an estimate of the theoretical probability parameter δ_g.

Recovery is summarized by an episode-level recovery-success indicator and post-attack time-to-recovery. Safety is summarized by the episode-level violation indicator and the minimum native margin. The shield-intervention ratio records the fraction of time steps at which the execution layer changes the authorized proposal.

E. Statistical Protocol

All headline comparisons use the same 20 matched seeds, $\text{seed} \in \{0, 1, \ldots, 19\}$. Continuous integrity, margin, and recovery metrics are compared with paired Wilcoxon signed-rank tests. For each metric, the six AEFC-Full-versus-alternative comparisons are corrected with Holm's procedure. Exact all-zero paired differences are assigned $p = 1$ before multiplicity correction, preventing a no-effect comparison from becoming spuriously significant through non-finite-value handling. The generated statistics also report paired median differences. Episode-level binary proportions use Wilson 95% confidence intervals; continuous table entries use mean 95% confidence intervals across the 20 matched seeds.

F. Trace-Generated Results

The mechanism-level and native result sections below are imported directly from generated LaTeX files. The manuscript build therefore fails if the committed result files are absent rather than silently substituting unverified numerical values.

G. Mechanism-Level 20-Seed Joint-Corruption Results

The executable bounded-authority pipeline was evaluated under the joint physical/knowledge attack on the released reduced-order CPS integration backend. This backend exercises PATBU, the risk gate, trust-gated adaptation, Byzantine-resilient coordination, the robust execution shield, and post-shield control corruption. It is used to validate mechanism composition and is not presented as an IEEE 39-bus result. All values below are generated from 20 matched seeds and immutable raw JSONL traces tied to one commit and configuration.

For the joint attack, AEFC-Full yields episode safety-violation rate 0.0000 (95% CI [0.0000, 0.1611]), versus 1.0000 ([0.8389, 1.0000]) for FedRL.

For the joint attack, AEFC-Full yields false-authorization rate 0.0000 (95% CI [0.0000, 0.0000]), versus 1.0000 ([1.0000, 1.0000]) for FedRL.

For the joint attack, AEFC-Full yields maximum belief distortion 0.4813 (95% CI [0.4788, 0.4838]), versus 1.2896 ([1.2029, 1.3762]) for FedRL.

For the joint attack, AEFC-Full yields maximum deployed-policy drift 0.0360 (95% CI [0.0337, 0.0384]), versus 3.4224 ([3.0594, 3.7855]) for FedRL.

For the joint attack, AEFC-Full yields recovery success rate 1.0000 (95% CI [0.8389, 1.0000]), versus 0.9500 ([0.7639, 0.9911]) for FedRL.

For the joint attack, AEFC-Full yields post-attack recovery time 4.3000 (95% CI [3.8481, 4.7519]), versus 40.9000 ([38.6839, 43.1161]) for FedRL.

The ablations expose the intended layer separation. Removing the shield causes physical safety violations despite small policy drift, whereas removing trust-gated adaptation sharply increases false authorization and policy drift while the shield can still preserve the physical invariant. Removing PATBU increases belief distortion. These observations are mechanism-level evidence for the compositional design, not a substitute for native IEEE 39-bus validation.

Paired Wilcoxon tests for the continuous integrity and recovery metrics are computed per seed and corrected across the six alternative methods with Holm's procedure. The corresponding machine-readable statistics are released with the artifact.

H. Native IEEE 39-Bus 20-Seed Joint-Corruption Results

The native experiment uses the MathWorks R2024b `IEEE39BusSystem` plant. Four distinct generator-local `Pref` actuation edges are paired with physical rotor-speed observations from the same generators and accepted only after a full-rank finite-difference causal probe. The same seven methods, four corruption cells, and 20 matched seeds are then executed with control corruption injected downstream of the shield and upstream of the validated native actuator edges.

Under joint corruption, AEFC-Full yields episode safety-violation rate 0.0000 (95% CI [0.0000, 0.1611]), versus 0.0000 ([0.0000, 0.1611]) for FedRL.

Under joint corruption, AEFC-Full yields minimum native safety margin 0.9857 (95% CI [0.9837, 0.9878]), versus 0.9750 ([0.9716, 0.9783]) for FedRL.

Under joint corruption, AEFC-Full yields false-authorization rate 0.0000 (95% CI [0.0000, 0.0000]), versus 0.0000 ([0.0000, 0.0000]) for FedRL.

Under joint corruption, AEFC-Full yields maximum belief distortion 0.0873 (95% CI [0.0834, 0.0913]), versus 0.4978 ([0.4918, 0.5038]) for FedRL.

Under joint corruption, AEFC-Full yields maximum deployed-policy drift 0.0230 (95% CI [0.0211, 0.0250]), versus 0.1407 ([0.1315, 0.1499]) for FedRL.

Under joint corruption, AEFC-Full yields recovery success rate 1.0000 (95% CI [0.8389, 1.0000]), versus 1.0000 ([0.8389, 1.0000]) for FedRL.

Under joint corruption, AEFC-Full yields post-attack recovery time 0.0167 (95% CI [0.0167, 0.0167]), versus 0.0167 ([0.0167, 0.0167]) for FedRL.

For AEFC-Full versus FedRL, the paired median differences are -0.4095 for maximum belief distortion, -0.1140 for maximum deployed-policy drift, and 0.0102 for minimum native safety margin; all three comparisons remain significant after Holm correction ($p_{\text{adj}} = 1.14 \times 10^{-5}$ for each comparison).

Continuous integrity, safety-margin, and recovery metrics use paired Wilcoxon tests across matched seeds with Holm correction across the six alternative methods; exact zero paired differences are assigned $p = 1$ before Holm correction. The corresponding machine-readable statistics are released with the artifact.

TABLE II
MECHANISM-LEVEL JOINT-CORRUPTION RESULTS OVER 20 MATCHED SEEDS. VALUES ARE MEANS WITH 95% CONFIDENCE INTERVALS; WILSON INTERVALS ARE USED FOR BINARY RATES. THIS TABLE IS NOT AN IEEE 39-BUS RESULT.

Method	SVR ↓	FAR ↓	Belief dist. ↓	Policy drift ↓	Rec. success ↑	Rec. time ↓
FedRL	1.000 [0.839, 1.000]	1.000 [1.000, 1.000]	1.290 [1.203, 1.376]	3.422 [3.059, 3.785]	0.950 [0.764, 0.991]	40.900 [38.684, 43.116]
RobustAgg	0.050 [0.009, 0.236]	0.050 [0.000, 0.148]	0.819 [0.808, 0.829]	0.238 [0.226, 0.250]	1.000 [0.839, 1.000]	9.050 [5.663, 12.437]
AEFC-no-PATBU	0.000 [0.000, 0.161]	0.000 [0.000, 0.000]	0.861 [0.849, 0.872]	0.044 [0.042, 0.047]	1.000 [0.839, 1.000]	1.850 [1.556, 2.144]
AEFC-no-Gate	0.000 [0.000, 0.161]	0.000 [0.000, 0.000]	0.468 [0.464, 0.472]	0.047 [0.044, 0.050]	1.000 [0.839, 1.000]	3.500 [3.234, 3.766]
AEFC-no-TGOPA	0.000 [0.000, 0.161]	0.994 [0.984, 1.000]	0.483 [0.480, 0.486]	0.543 [0.524, 0.562]	1.000 [0.839, 1.000]	10.400 [7.064, 13.736]
AEFC-no-Shield	1.000 [0.839, 1.000]	0.014 [0.000, 0.029]	0.499 [0.495, 0.503]	0.036 [0.033, 0.039]	1.000 [0.839, 1.000]	4.500 [3.940, 5.060]
AEFC-Full	0.000 [0.000, 0.161]	0.000 [0.000, 0.000]	0.481 [0.479, 0.484]	0.036 [0.034, 0.038]	1.000 [0.839, 1.000]	4.300 [3.848, 4.752]

TABLE III
NATIVE MATHWORKS IEEE 39-BUS JOINT-CORRUPTION RESULTS OVER 20 MATCHED SEEDS. VALUES ARE MEANS WITH 95% CONFIDENCE INTERVALS; WILSON INTERVALS ARE USED FOR BINARY RATES.

Method	SVR ↓	Min. margin ↑	FAR ↓	Belief dist. ↓	Policy drift ↓	Rec. success ↑	Rec. time ↓
FedRL	0.000 [0.000, 0.161]	0.975 [0.972, 0.978]	0.000 [0.000, 0.000]	0.498 [0.492, 0.504]	0.141 [0.132, 0.150]	1.000 [0.839, 1.000]	0.017 [0.017, 0.017]
RobustAgg	0.000 [0.000, 0.161]	0.973 [0.971, 0.976]	0.000 [0.000, 0.000]	0.494 [0.490, 0.499]	0.111 [0.104, 0.118]	1.000 [0.839, 1.000]	0.017 [0.017, 0.017]
AEFC-no-PATBU	0.000 [0.000, 0.161]	0.987 [0.985, 0.989]	0.000 [0.000, 0.000]	0.498 [0.493, 0.503]	0.018 [0.017, 0.020]	1.000 [0.839, 1.000]	0.017 [0.017, 0.017]
AEFC-no-Gate	0.000 [0.000, 0.161]	0.986 [0.984, 0.988]	0.000 [0.000, 0.000]	0.087 [0.083, 0.091]	0.023 [0.021, 0.025]	1.000 [0.839, 1.000]	0.017 [0.017, 0.017]
AEFC-no-TGOPA	0.000 [0.000, 0.161]	0.973 [0.971, 0.976]	0.000 [0.000, 0.000]	0.087 [0.083, 0.090]	0.111 [0.104, 0.118]	1.000 [0.839, 1.000]	0.017 [0.017, 0.017]
AEFC-no-Shield	0.000 [0.000, 0.161]	0.986 [0.984, 0.988]	0.000 [0.000, 0.000]	0.087 [0.083, 0.091]	0.023 [0.021, 0.025]	1.000 [0.839, 1.000]	0.017 [0.017, 0.017]
AEFC-Full	0.000 [0.000, 0.161]	0.986 [0.984, 0.988]	0.000 [0.000, 0.000]	0.087 [0.083, 0.091]	0.023 [0.021, 0.025]	1.000 [0.839, 1.000]	0.017 [0.017, 0.017]

V. SECURITY VALIDATION AND LIMITATIONS

A. Claim-to-Evidence Correspondence

The formal guarantees and the executable evidence serve different purposes and are therefore interpreted separately.

G1: bounded belief distortion. The mechanism-level traces record the clean counterfactual belief and the attacked belief, while the native controller records the resulting distortion diagnostic. These measurements test whether the implemented belief mechanism behaves consistently with the intended contraction structure. They do not replace the analytical assumption that the belief operator is contractive or establish a worst-case value of the bound in (16).

G2: bounded false authorization. The experiments report an operational false-authorization rate based on the logged gate decision and a fixed dangerous-proposal indicator. As defined in the experimental section, this indicator is backend-specific. The empirical FAR therefore measures authorization behavior under the released attack configurations; it is not a direct estimate of the theorem parameter δ_g, whose interpretation still depends on calibration or deterministic outer coverage of the risk bound.

G3–G4: bounded adaptation and post-poisoning correction. The mechanism-level backend records candidate/deployed-policy hashes and the deployed-policy movement, while the native path records adaptation decisions and the same movement metric. The released implementation does not provide a cryptographically certified rollback service. Consequently, the manuscript claims finite ε-correction only under the stated post-attack contraction condition and does not claim finite exact return of the policy parameters after poisoning.

G5: physical forward invariance. G5 is an analytical result for a feasible robust barrier-constrained execution layer under uncertainty-set and attack-set coverage. The native R2024b controller used in the 560-episode experiment implements a bounded sensitivity-based safety projection constructed from the validated 4×4 finite-difference mapping; it is not a full online CBF-QP solver. Accordingly, the native experiment reports observed safety-envelope violations and continuous native margins, but it is not presented as an empirical proof of the CBF forward-invariance theorem. This separation prevents an implementation-level safety observation from being conflated with the stronger analytical guarantee.

B. Byzantine Aggregation and Sparse-Support Scope

Byzantine clients may fabricate update values, timestamps, sparse indices, advantages, prototypes, confidence values, and other self-reported metadata. Such quantities are therefore never treated as authentication signals. The mechanism-level path uses a common-support robust aggregation interface and records the aggregate deviation from the honest center. The four-dimensional native controller receives complete local update vectors and applies robust shared-support selection before trimmed aggregation; its native results are not used to make a communication-compression claim.

For deterministic Top-k compression, the implementation-level audit is based on

$$\frac{\|z_i^t - C_k(z_i^t)\|_2^2}{\|z_i^t\|_2^2} \le 1 - \frac{k}{d}. \tag{54}$$

This result characterizes compression error only. It does not authenticate a client, provide Byzantine provenance, or establish privacy.

C. Native IEEE 39-Bus Evidence Boundary

The authoritative native evidence is tied to the successful MathWorks R2024b workflow at commit `cc315ff522c90ce6adc09df3020472dd9ad820ea`. The workflow validates four generator-local rotor-speed/`Pref` pairs with a full-rank sensitivity matrix, executes 560 native episodes at $1/60$ s, uses orientation-safe `SimulationOutput` extraction with per-episode

semantic guards, and releases immutable per-seed traces and summaries. The reduced-order 20-seed experiment remains a mechanism-integration test and is not used as an IEEE 39-bus substitute.

The native evidence is deliberately interpreted within the configured attack envelope. Under joint corruption, all seven methods exhibit zero episode-level safety violations and 100% recovery success across 20 matched seeds. The native experiment therefore does not support a claim that AEFC uniquely improves these binary outcomes in this operating regime. Its measured advantage is concentrated in belief integrity, deployed-policy movement, and the continuous native safety margin. Relative to FedRL, AEFC-Full reduces maximum belief distortion and policy movement and increases the minimum native margin, with the corresponding paired comparisons remaining significant after Holm correction. Exact zero paired differences are assigned $p = 1$ before multiplicity correction.

The physical theorem remains conditional on uncertainty-set coverage and robust-shield feasibility. An adversary capable of imposing arbitrary unbounded downstream actuation lies outside the guarantee. IEEE 33-bus, IEEE 118-bus, and 8500-node quantitative results are not part of the released evidence used for the claims in this manuscript.

D. Provenance and Privacy Scope

Provenance authenticity requires a cryptographic authentication, trusted-time, or attestation mechanism appropriate to the deployment; a client-provided source identifier alone is not authenticated provenance. The current artifact models provenance validity but does not claim that this systems problem is solved by metadata fields themselves.

The AEFC security claim concerns integrity and physical safety rather than formal confidentiality. Data locality and sparse updates do not constitute differential privacy. No (ε, δ) privacy guarantee is claimed, and membership-inference behavior is not used as evidence for the bounded-authority security results.

These boundaries are part of the security model rather than post hoc qualifications. They identify which claims follow from analysis, which are supported by the released traces, and which remain deployment assumptions.

VI. CONCLUSION

This paper reframes federated CPS recovery as a bounded-authority security problem. The central challenge is not merely that physical observations and control channels may be attacked or that federated learning may be poisoned; it is that corrupted recovery knowledge can propagate through belief formation, action authorization, online adaptation, and physical execution. AEFC therefore treats federated recovery knowledge as untrusted evidence rather than trusted controller state.

The design separates five guarantees. Provenance-aware temporal processing bounds belief distortion; the risk-aware gate bounds false authorization; trust-gated deployment bounds policy movement and supports explicit post-poisoning correction conditions; Byzantine-resilient coordination limits the influence of malicious updates; and the robust physical execution layer preserves the safe set under the stated coverage and feasibility assumptions. This composition gives the federation a bounded security role rather than relying solely on a final safety filter.

Heterogeneous adapters, recovery-state alignment, and deterministic Top-k remain useful for learning under subsystem diversity and communication constraints, but they are not treated as trust or privacy mechanisms. Likewise, the paper does not infer a formal privacy guarantee from data locality or sparse communication.

The experimental evidence follows the same separation. The released mechanism-level backend validates composition behavior, while the orientation-safe native MathWorks R2024b IEEE 39-bus evaluation executes seven methods, four corruption cells, and 20 matched seeds per cell, yielding 560 immutable native episodes. Under joint corruption, all compared methods remain free of episode-level safety violations and recover within the observed horizon, so no binary safety advantage is attributed to AEFC in this operating regime. The native evidence instead shows that AEFC-Full substantially reduces belief distortion and deployed-policy movement relative to FedRL while increasing the minimum native safety margin; the corresponding paired comparisons remain significant after Holm correction. These results are generated directly from validated generator-local rotor-speed/`Pref` interfaces and immutable per-seed traces.

Taken together, the analysis and evidence support a scoped conclusion: federated recovery knowledge may be adversarial, yet its authority over a safety-critical CPS can be bounded before it reaches the plant. The formal safety guarantee remains conditional on uncertainty-set coverage, bounded downstream corruption, and shield feasibility, while the native IEEE 39-bus results demonstrate measurable integrity benefits without overstating binary outcomes that are identical across the compared methods.

ACKNOWLEDGMENTS

This work was partially supported by the National Natural Science Foundation of China (Grant Nos. 62502154, 62472151, 62576051, and U23A20317), the Creative Research Groups Program of the National Natural Science Foundation of China (Grant No. 62321003), the Postdoctoral Fellowship Program of the China Postdoctoral Science Foundation (Grant No. GZC20252182), the Natural Science Foundation of Hunan Province (Grant No. 2026JJ60224), the Provincial Key R&D Program of Hunan (Grant No. 2024AQ2025), the Yuelu Mountain Industrial Innovation Center Project (Grant No. 2023YCII0118), and the Scientific Research Fund of the Hunan Provincial Education Department (Grant No. 22A0592).

REFERENCES

[1] M. Segovia-Ferreira, J. Rubio-Hernan, A. Cavalli, and J. Garcia-Alfaro, "A survey on cyber-resilience approaches for cyber-physical systems," *ACM Computing Surveys*, vol. 56, no. 8, pp. 1–37, 2024.

[2] S. O'Toole and H. Mehrpouyan, "Towards cyber-physical representation and cyber-resilience against attack and failure within a hydraulic network simulation toolkit," in *2024 IEEE Security and Privacy Workshops (SPW)*. IEEE, 2024, pp. 246–252.

[3] P. Lu, L. Zhang, M. Liu, K. Sridhar, O. Sokolsky, F. Kong, and I. Lee, "Recovery from adversarial attacks in cyber-physical systems: Shallow, deep, and exploratory works," *ACM Computing Surveys*, vol. 56, no. 8, pp. 1–31, 2024.

[4] H. Yuan, Y. Yuan, Y. Zhong, and Y. Xia, "Incomplete information-based resilient strategy design for cyber-physical systems under stochastic communication protocol," *IEEE Transactions on Industrial Electronics*, vol. 71, no. 11, pp. 14 967–14 976, 2024.

[5] S. Kim, K.-J. Park, and C. Lu, "A survey on network security for cyber–physical systems: From threats to resilient design," *IEEE Communications Surveys & Tutorials*, vol. 24, no. 3, pp. 1534–1573, 2022.

[6] L. Niu, D. Sahabandu, A. Clark, and R. Poovendran, "Verifying safety for resilient cyber-physical systems via reactive software restart," in *2022 ACM/IEEE 13th International Conference on Cyber-Physical Systems (ICCPS)*. IEEE, 2022, pp. 104–115.

[7] R. Yu, X. Zhao, S. Lu, Y.-H. Chen, and C. Li, "Intelligent game-theoretic approach for resilient robust control design of cyber-physical systems: application to intelligent transportation systems," *IEEE Transactions on Intelligent Transportation Systems*, vol. 25, no. 11, pp. 16 072–16 083, 2024.

[8] L. Xin and Z.-Q. Long, "A learning-based passive resilient controller for cyber-physical systems: Countering stealthy deception attacks and complete loss of actuators control authority," *IEEE/CAA Journal of Automatica Sinica*, vol. 12, no. 7, pp. 1368–1380, 2025.

[9] K. Zhong, Z. Yang, S. Yu, and K. Li, "Deep reinforcement learning-based multi-layer cascaded resilient recovery for cyber-physical systems," *IEEE Transactions on Services Computing*, 2024.

[10] F. O. Olowononi, D. B. Rawat, and C. Liu, "Resilient machine learning for networked cyber physical systems: A survey for machine learning security to securing machine learning for cps," *IEEE Communications Surveys & Tutorials*, vol. 23, no. 1, pp. 524–552, 2020.

[11] T. T. Nguyen and V. J. Reddi, "Deep reinforcement learning for cyber security," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 34, no. 8, pp. 3779–3795, 2021.

[12] D. Zhang, G. Feng, Y. Shi, and D. Srinivasan, "Physical safety and cyber security analysis of multi-agent systems: A survey of recent advances," *IEEE/CAA Journal of Automatica Sinica*, vol. 8, no. 2, pp. 319–333, 2021.

[13] W. Jin, H. Du, B. Zhao, X. Tian, B. Shi, and G. Yang, "A comprehensive survey on multi-agent cooperative decision-making: Scenarios, approaches, challenges and perspectives," *arXiv preprint arXiv:2503.13415*, 2025.

[14] Y. Yu, W. Yang, W. Ding, and J. Zhou, "Reinforcement learning solution for cyber-physical systems security against replay attacks," *IEEE Transactions on Information Forensics and Security*, vol. 18, pp. 2583–2595, 2023.

[15] P. Su, Z. Lu, and D. Chen, "Combining self-organizing map with reinforcement learning for multivariate time series anomaly detection," in *2023 IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, 2023, pp. 1964–1969.

[16] X. Yang, E. Howley, and M. Schukat, "Adt: Time series anomaly detection for cyber-physical systems via deep reinforcement learning," *Computers & Security*, vol. 141, p. 103825, 2024.

[17] M. Liu, P. Lu, X. Chen, O. Sokolsky, I. Lee, and F. Kong, "Deadline-safe reach-avoid control synthesis for cyber-physical systems with reinforcement learning," in *2024 IEEE Real-Time Systems Symposium (RTSS)*. IEEE, 2024, pp. 96–108.

[18] Y. Wang, W. Xie, Z. Ding, Q. Yang, Y. Lin, and Y. Zhang, "Multi-agent graph reinforcement learning based cross-layer routing for mobile ad-hoc network," in *International Conference on Internet of Things, Communication and Intelligent Technology*. Springer, 2023, pp. 354–364.

[19] P. Kairouz, H. B. McMahan, B. Avent, A. Bellet, M. Bennis, A. N. Bhagoji, K. Bonawitz, Z. Charles, G. Cormode, R. Cummings *et al.*, "Advances and open problems in federated learning," *Foundations and trends® in machine learning*, vol. 14, no. 1–2, pp. 1–210, 2021.

[20] T. Li, A. K. Sahu, A. Talwalkar, and V. Smith, "Federated learning: Challenges, methods, and future directions," *IEEE signal processing magazine*, vol. 37, no. 3, pp. 50–60, 2020.

[21] R. Zhao, Z. Wang, D. Liu, S. Lv, and L. T. Yang, "Fedara: Resource-adaptive low-rank personalized federated learning via anchor-driven representation alignment on heterogeneous edge devices," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026.

[22] H. Park, M. Joe, M. Kim, K. Im, and M. Kwon, "Fed-ade: Adaptive learning rate for federated post-adaptation under distribution shift," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026.

[23] H. Q. Le, L. X. Nguyen, Y. Qiao, S. T. Kim, E.-N. Huh, and C. S. Hong, "Feddap: Domain-aware prototype learning for federated learning under domain shift," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026.

[24] H. Wang, J. Shen, J. Yan, and G. Pang, "Domain-skewed federated learning with feature decoupling and calibration," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026.

[25] N. Jia, Z. Qu, B. Ye, Y. Wang, S. Hu, and S. Guo, "A comprehensive survey on communication-efficient federated learning in mobile edge environments," *IEEE Communications Surveys & Tutorials*, 2025.

[26] S. Zheng, L. Zhang, Y. Dong, X. Li, X. Wang, and X. Chen, "Toward communication-efficient over-the-air federated learning: Synergistic compression for uplink and downlink transmission," *IEEE Internet of Things Journal*, pp. 1–1, 2025.

[27] Y. Miao, Z. Liu, H. Li, K.-K. R. Choo, and R. H. Deng, "Efficient and secure federated learning against backdoor attacks," *IEEE Transactions on Dependable and Secure Computing*, vol. 21, no. 5, pp. 4619–4636, 2024.

[28] Y. Miao, Z. Liu, X. Li, M. Li, H. Li, K.-K. R. Choo, and R. H. Deng, "Privacy-preserving byzantine-robust federated learning via blockchain systems," *IEEE Transactions on Dependable and Secure Computing*, vol. 21, no. 4, pp. 2361–2375, 2024.

[29] X. Mu, K. Cheng, Y. Shen, X. Li, Z. Chang, T. Zhang, and X. Ma, "Feddmc: Efficient and robust federated learning via detecting malicious clients," *IEEE Transactions on Dependable and Secure Computing*, vol. 21, no. 6, pp. 5259–5274, 2024.

[30] J. Zhao, H. Zhu, F. Wang, R. Lu, and H. Li, "Sxgb: Secure and efficient vertical federated xgboost via trusted execution environments," *IEEE Transactions on Dependable and Secure Computing*, vol. 23, no. 2, pp. 2275–2288, 2026.

[31] F. Kato, Y. Cao, and M. Yoshikawa, "Olive: Oblivious and differentially private federated learning on trusted execution environment," *Proceedings of the VLDB Endowment*, vol. 16, no. 10, pp. 2404–2417, 2023.

[32] J. Liu, N. Xi, Y. Miao, and J. Liu, "Dp-fedadamw: An efficient optimizer for differentially private federated large models," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026.

[33] Y. Zhao, X. Deng, H. Xu, Z. Qiu, X. Hu, S. You, Y. Chen, C. Xu, and X. Su, "Fedmop: Achieving enhanced privacy and performance in federated learning via momentum orthogonal projection," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026.

[34] K. K. Patel, B. Jiang, A. F. M. M. Kabir, W. Zhang, D. Zou, and L. Wang, "Personalized federated training of diffusion models with privacy guarantees," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026.

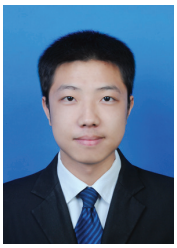
[35] J. Qi, Q. Zhou, L. Lei, and K. Zheng, "Federated reinforcement learning: Techniques, applications, and open challenges," *arXiv preprint arXiv:2108.11887*, 2021.

[36] C. Feng, A. H. Celdrán, P. M. S. Sánchez, J. Kreischer, J. v. d. Assen, G. Bovet, G. M. Pérez, and B. Stiller, "Cyberforce: A federated reinforcement learning framework for malware mitigation," *IEEE Transactions on Dependable and Secure Computing*, vol. 22, no. 4, pp. 4398–4411, 2025.

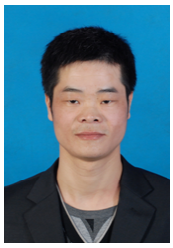
[37] T. Shaik, X. Tao, L. Li, H. Xie, T. Cai, X. Zhu, and Q. Li, "Framu: Attention-based machine unlearning using federated reinforcement learning," *IEEE Transactions on Knowledge and Data Engineering*, vol. 36, no. 10, pp. 5153–5167, 2024.

[38] L. Yang, S. Guo, C.-K. Tham, M. Li, G. Liu, and P. Zhou, "Prem-fediov: A novel federated reinforcement learning framework for predictive maintenance in iov," *IEEE Transactions on Mobile Computing*, vol. 23, no. 12, pp. 11 954–11 970, 2024.

[39] H. Qi, W. Ma, and B. Zhao, "Federated heterogeneous multi-agent deep reinforcement learning-based attack resilience scheduling for heterogeneous multi-integrated energy system," *Energy Reports*, vol. 14, pp. 116–127, 2025.



Kai Zhong received the PhD degree in computer science from Hunan University, Changsha, China, in 2024. He is currently a postdoctoral researcher at Hunan University. His research interests include cybersecurity, industrial control systems, and edge-cloud computing.



Zhibang Yang received the PhD degree in computer science from Hunan University, Changsha, China, in 2012. He is currently an associate professor at Changsha University. His research interests include cybersecurity, industrial control systems, and edge-cloud computing.



Yumeng Liu received the PhD degree from the Institute of Software, Chinese Academy of Sciences, Beijing, China, in 2023. He is a senior engineer at the Institute of Software, Chinese Academy of Sciences. His research interests include data mining and database technology.



Siyang Yu received the PhD degree in computer science and technology from Hunan University, China, in 2017. He is currently a faculty member at Hunan University of Finance and Economics. His research interests include the industrial Internet of Things, anomaly analysis, and intrusion detection.



Kenli Li received the PhD degree in computer science from Huazhong University of Science and Technology, China, in 2003. He is currently a Cheung Kong professor of computer science and technology with Hunan University and the dean of the College of Computer Science and Electronic Engineering, Hunan University. His research interests include parallel and distributed processing, high-performance computing, and big data management. He has published more than 250 research papers in international conferences and journals such as TC, TPDS, TCC, ICPP, ICDCS, etc. He serves on the editorial board of the IEEE-TC. He is an outstanding member of the CCF and a senior member of the IEEE.